

DATA CLEANING LOG

Credit Bureau Dataset — Portfolio Project

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Tool: Python (pandas)
Date: June 2025

Raw Dataset	credit_bureau_dataset.csv 10,150 rows 22 columns
Cleaned Dataset	credit_bureau_dataset_clean.csv 10,000 rows 22 columns
Rows Removed	150 (duplicate records)
Issues Fixed	5 issue types addressed across 6 columns

Raw Dataset Summary

The raw dataset was generated as a synthetic credit bureau dataset containing intentional data quality issues for portfolio cleaning practice. Before any cleaning, the dataset had the following profile:

Dimension	Raw	Cleaned	Change
Total Rows	10,150	10,000	-150
Total Columns	22	22	None
Duplicate Rows	150	0	-150 removed
Missing Values	1,545	0	All resolved
Inconsistent Cols	2	0	Gender, Employment_Status standardised

Missing Values by Column (Raw)

Column	Data Type	Missing Count	Missing %	Resolution
Annual_Income	Float	290	2.86%	Median imputation
Credit_Score	Integer	375	3.69%	Median imputation
Employment_Status	String	331	3.26%	Filled as 'Unknown'
Monthly_Payment	Float	255	2.51%	Median imputation
Credit_Utilization_Rate	Float	294	2.90%	Median imputation

SECTION 2 ISSUES IDENTIFIED & ACTIONS TAKEN

Issue 1 — Duplicate Records

Detail	Value
Columns Affected	All columns
Records Affected	150 fully duplicated rows
Detection Method	df.duplicated() and df.duplicated(subset='Customer_ID')
Action Taken	Removed all 150 duplicate rows using df.drop_duplicates()
Rows After	10,000
Justification	Duplicates inflate record counts and skew all aggregations

Python used:

```
df = df.drop_duplicates() # removed 150 rows
```

Issue 2 — Missing Values in Numeric Columns

Four numeric columns contained missing values. The **median** was chosen over the mean because Annual_Income and Monthly_Payment contain outliers that would skew a mean-based imputation. Using the median preserves the central tendency of the data without being distorted by extreme values.

Column	Nulls	Median Used	Justification
Annual_Income	290	NGN 622,788	Outliers present — median more robust
Credit_Score	375	607	Median and mean closely aligned; median preferred
Monthly_Payment	255	NGN 36,337	Outliers present — mean (NGN 158k) far exceeds median
Credit_Utilization_Rate	294	0.26	Stable central value within 0-1 range

Python used:

```
numeric_cols = ['Annual_Income', 'Credit_Score', 'Monthly_Payment', 'Credit_Utilization_Rate']  
for col in numeric_cols: df[col].fillna(df[col].median(), inplace=True)
```

Issue 3 — Missing Values in Employment_Status (Categorical)

Employment_Status had 331 missing values. Unlike numeric columns, filling a categorical column with the mode would artificially inflate the 'Employed' category. Instead, nulls were filled with **'Unknown'** to preserve transparency about data gaps without introducing bias.

Decision	Detail
Nulls Found	331 rows
Options Considered	Mode imputation ('Employed') OR label as 'Unknown'
Choice Made	Filled with 'Unknown'
Reason	Avoids inflating any one category; keeps gap visible in analysis

Python used:

```
df['Employment_Status'].fillna('Unknown', inplace=True)
```

Issue 4 — Inconsistent Formatting in Gender

The Gender column contained 8 different representations of only 2 values, including mixed case, abbreviations, and varying capitalisation.

Raw Value	Count	Standardised To
FEMALE	1,324	Female
MALE	1,310	Male
Female	1,294	Female
male	1,283	Male
Male	1,264	Male
F	1,246	Female
M	1,243	Male
female	1,186	Female

Python used:

```
df['Gender'] = df['Gender'].str.strip() df['Gender'] = df['Gender'].replace({'M': 'Male', 'F': 'Female', 'male': 'Male', 'female': 'Female', 'MALE': 'Male', 'FEMALE': 'Female'})
```

Issue 5 — Inconsistent Formatting in Employment_Status

Employment_Status had 9 raw variants representing only 5 actual categories, caused by mixed case, spacing differences, and abbreviation inconsistency.

Raw Value	Standardised To
self employed	Self-Employed
Self employed	Self-Employed
Self-Employed	Self-Employed
EMPLOYED	Employed
employed	Employed
Employed	Employed
Unemployed	Unemployed
Student	Student
Retired	Retired
(null)	Unknown

Python used:

```
df['Employment_Status'] = df['Employment_Status'].str.strip() df['Employment_Status'] = df['Employment_Status'].replace({'employed': 'Employed', 'EMPLOYED': 'Employed', 'self employed': 'Self-Employed', 'Self employed': 'Self-Employed' }) df['Employment_Status'].fillna('Unknown', inplace=True)
```

SECTION 3 KNOWN LIMITATIONS & ANALYST NOTES

Items Noted But Not Changed

The following observations were made during the cleaning process but were deliberately left unchanged, with reasoning documented below.

Item	Observation	Decision & Reason
High Annual Income	~1% of income values are 8–20x above average (up to NGN 35.9M)	Retained. High earners — executives, business owners — are realistic in Nigeria. These are not data errors.
Retired Income < NGN 840k	147 retired customers earn below the minimum wage equivalent (NGN 70k/month)	Flagged but retained. Some retirees may lack formal pension. Worth filtering in specific analyses.
High Default Rate (33%)	Overall default rate is 33%, higher than a typical real-world portfolio (~5–10%)	Expected. The dataset was designed with an inflated default rate to support richer analytical scenarios.
Unknown Employment	328 rows now labelled 'Unknown' in Employment_Status	Intentional. Preserves visibility of the data gap without fabricating a category.

Cleaning Decisions Summary

Median over mean for imputation

Annual_Income and Monthly_Payment contain outliers that inflate the mean significantly. The median is resistant to skew and better represents the typical customer value.

'Unknown' over mode for categorical nulls

Filling Employment_Status with 'Employed' (the mode) would inflate that category by ~10%. 'Unknown' is honest and keeps the gap visible.

Raw file preserved

The original messy CSV was kept intact. The cleaned version is a separate file. This is best practice — never overwrite raw data.

No outlier removal on income

Extreme incomes are plausible for high earners in Nigeria. Removing them would erase a real customer segment and reduce analytical richness.

Cleaning completed by Oluwabusola Oyenuka | Tool: Python (pandas) | June 2025